

ML - Learning

Linear Regression 2 independent variables

$$a = \bar{Y} - b_1 \bar{X}_1 - b_2 \bar{X}_2$$

TP $\rightarrow$ Actual +ve
Pred +ve

TN $\rightarrow$ Actual -ve
Pred -ve

FP $\rightarrow$ Actual -ve
Pred +ve

FN $\rightarrow$ Actual +ve
Pred -ve

Confusion Matrix For Slide-6

	Pred	
	+ve	-ve
Actual +ve	45	15
Actual -ve	8	32

$$\text{Accuracy} = \frac{TP + TN}{N} = \frac{45 + 32}{100} = 77\% \text{ } 0.77$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{45}{45 + 8} = 0.849$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{45}{45 + 15} = 0.75$$

$$F1 \text{ Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$= \frac{2 \times 0.849 \times 0.75}{0.849 + 0.75} = 0.79$$

$$\text{Type-1 Error} = \text{FP}$$

$$\text{Type-2 Error} = \text{FN}$$

Accuracy is same so we have to analyze the case. IF FP is dangerous for the model so we will calculate Precision. IF FN is vulnerable so we will calculate Recall.

Multi-Class Confusion Matrix:-

		Pred			
		A	M	G	Total
Actual	A	25	5	7	37
	M	4	32	2	38
	G	6	4	25	35
Total		35	41	34	110

$$\text{Accuracy} = \frac{25 + 32 + 25}{110} = 0.74$$

$$\text{Precision A} = \frac{25}{35} = 0.71$$

Precision For Different classes:-

$$Pre = \frac{\text{True Predictions of class}}{\text{Total Prediction of class}}$$

For the Precision of whole model:-

$$\text{Macro-Average} = \frac{0.71 + 0.78 + 0.73}{3} = 0.74$$

$$\text{Weighted Prec} = \frac{37}{110} (0.71) + \frac{38}{110} (0.78) + \frac{35}{110} (0.73)$$

$$\text{Recall}_A = \frac{T_A}{\text{Actual Apple}} = \frac{25}{37}$$

Same will be for Recall for whole model.

$$F1A = \frac{2 \times P_A \times R_A}{P_A + R_A}$$

Decision - TREE:-

In decision Tree, there are 3 Algo;

- (i) ID3 $\rightarrow$ Entropy & Information Gain
- (ii) CART $\rightarrow$ Gini Impurity
- (iii) C4.5 $\rightarrow$

$$\frac{20(0.6)}{\log_{20}(2)}$$

ID.3 Algorithm:-

$$\text{Entropy} = \frac{-P}{P+n} \log_2 \frac{P}{P+n} - \frac{n}{P+n} \log_2 \frac{n}{P+n}$$

→ cloudy = 4, Yes

~~Information~~ Entropy = 0 (Highly certain)

Entropy = 1 (Highly uncertain)

→ Rain = 4, Yes = 2 & No = 2

$$\text{Information}_{\text{att}} = \sum_{i=1}^n \frac{P_i + n_i}{N} \text{Ent}_i$$

$$\text{Gain}_{\text{att}} = \text{Ent}_{\text{Total}} - \text{Info}_{\text{att}}$$

Model.

Apply Decision Tree Algo. on Dataset (Slide-6)

~~Entropy~~ Entropy:-

Color	P	n	Ent
Red	2	1	From Formula
Green Blue	0	2	0
Green Blue	1	0	0

$$\text{Info color} = \sum_{i=1}^3 \frac{P_i + n_i}{N} \text{Ent}_i + \frac{P_B + n_B}{N} (\text{Ent}_B)$$

$$\text{Info color} = \frac{P_{\text{red}} + n_{\text{red}}}{N} (\text{Ent}_{\text{red}}) + \frac{P_B + n_B}{N} (\text{Ent}_B)$$

$$\text{Gain}_{\text{att}} = \text{Ent}_{\text{Total}} - \text{Info}_{\text{att}}$$

For Red entropy:-

$$-\frac{2}{3} \log_2 (2/3) - \frac{1}{3} \log_2 (1/3) = 0.918$$

$$\text{Info}_{\text{color}} = \frac{2+1}{6} (0.918) + \frac{0+2}{6} (0) + \frac{1+0}{6} (0)$$

$$= 0.45$$

$$\text{Gain}_{\text{color}} = 1 - 0.45 = 0.55$$

shape	P	n	Ent
Square	2	1	0.918
round	1	2	0.918

$$\text{Info}_{\text{shape}} = \frac{2+1}{6} (0.918) + \frac{1+2}{6} (0.918)$$

$$= 0.9$$

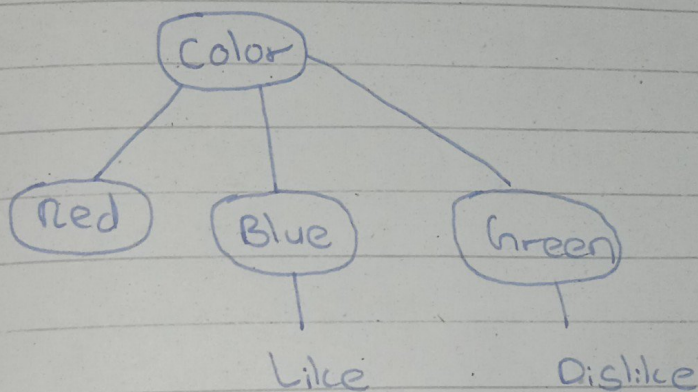
$$\text{Gain}_{\text{shape}} = 1 - 0.9 = 0.1$$

Size	P	n	Entropy
Large	3	1	0.81
Small	0	2	0

$$\text{Info}_{\text{shape}} = \frac{3+1}{6} (0.81) \\ = 0.54$$

$$\text{Gain}_{\text{shape}} = 1 - 0.54 = 0.46$$

Select the highest gain. Here is color



Now Red has both like and dislike So we calculate entropy selecting only Red data.

shape

Red	P	n	Ent
Square	1	0	0
Round	1	1	1

$$\text{Info}_{(\text{shape})} = \frac{2+1}{3} (1) = 0.66$$

Entropy of red
0.918

$$\text{Gain}(\text{shape}) = 1 - 0.66 = 0.251$$

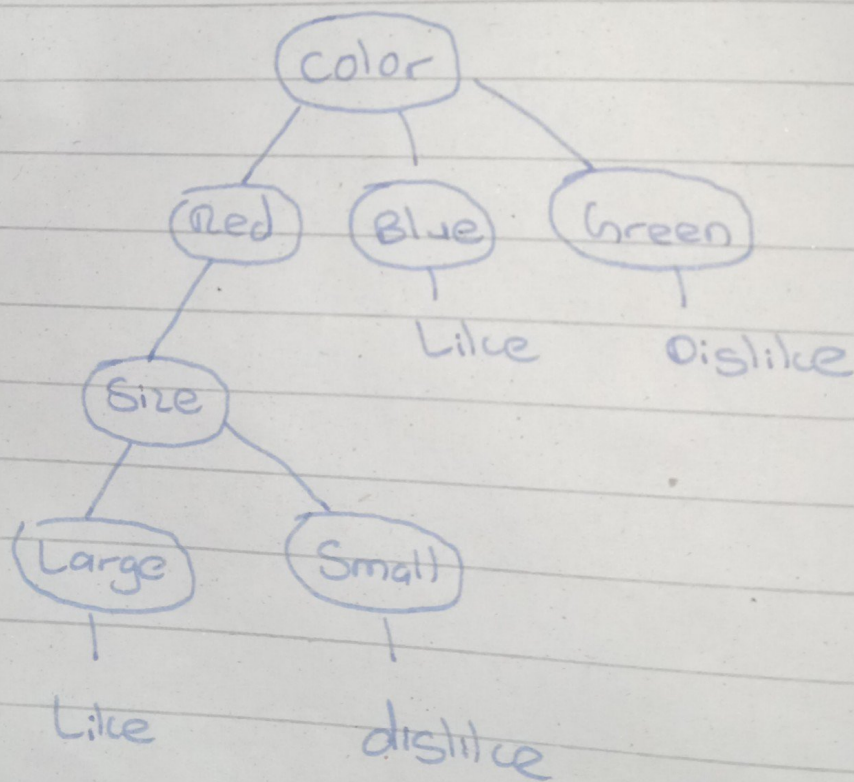
Now For Size

Size	P	n	Ent
Large	2	0	0
Small	0	1	0

Info will be zero

$$\text{Gain} = 0.918$$

So here Gain of Size is high



Naive Bayes Classifier:-

weather	N	P	$\rightarrow$ 2 (No. of Positive in Sunny) 9 (total Positives) Same for all
Sunny	$3/5$	$2/9$	
overcast	0	$4/9$	
Rain	$2/5$	$3/9$	

Temperature	N	P
Hot	$2/5$	$2/9$
cold	$1/5$	$3/9$
Mild	$2/5$	$4/9$

Humidity	N	P
High	$4/5$	$3/9$
Normal	$1/5$	$6/9$

windy	N	P
True	$3/5$	$3/9$
False	$2/5$	$6/9$

Probability (Positive) = $9/14$
 " (Negative) = $5/14$

Now one Test data is given in the Question

Rain | Hot | High | False | ?

$$P(X/\text{Yes}) = P(\text{rain}/\text{Yes}) \times P(\text{Hot}/\text{Yes}) \times P(\text{High}/\text{Yes}) \times P(\text{False}/\text{Yes})$$

$$= \frac{3}{9} \times \frac{2}{9} \times \frac{3}{9} \times \frac{6}{9} \times \frac{9}{24} = 0.03058$$

$$P(X/\text{No}) = P(\text{rain}/\text{No}) \times P(\text{Hot}/\text{No}) \times P(\text{High}/\text{No}) \times P(\text{False}/\text{No}) \times P(\text{No})$$

$$= \frac{2}{5} \times \frac{2}{5} \times \frac{4}{5} \times \frac{2}{5} \times \frac{5}{24} = 0.0382$$

Now $P(X/\text{No}) > P(X/\text{Yes})$

So it will be no play.

Icon Algorithm From AI Notes.

Question

Support Vector Machine Algorithm

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High / Yes

False / Yes / No

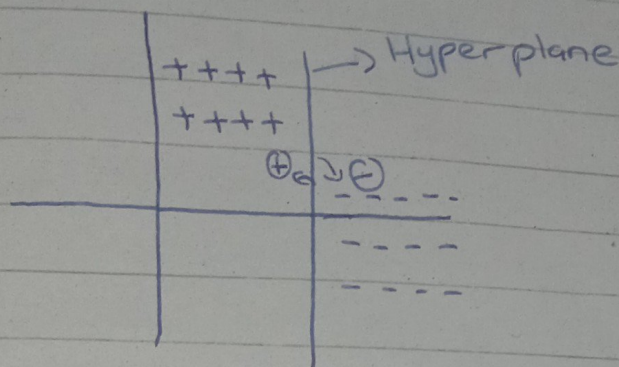
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1058

h / No

P / No

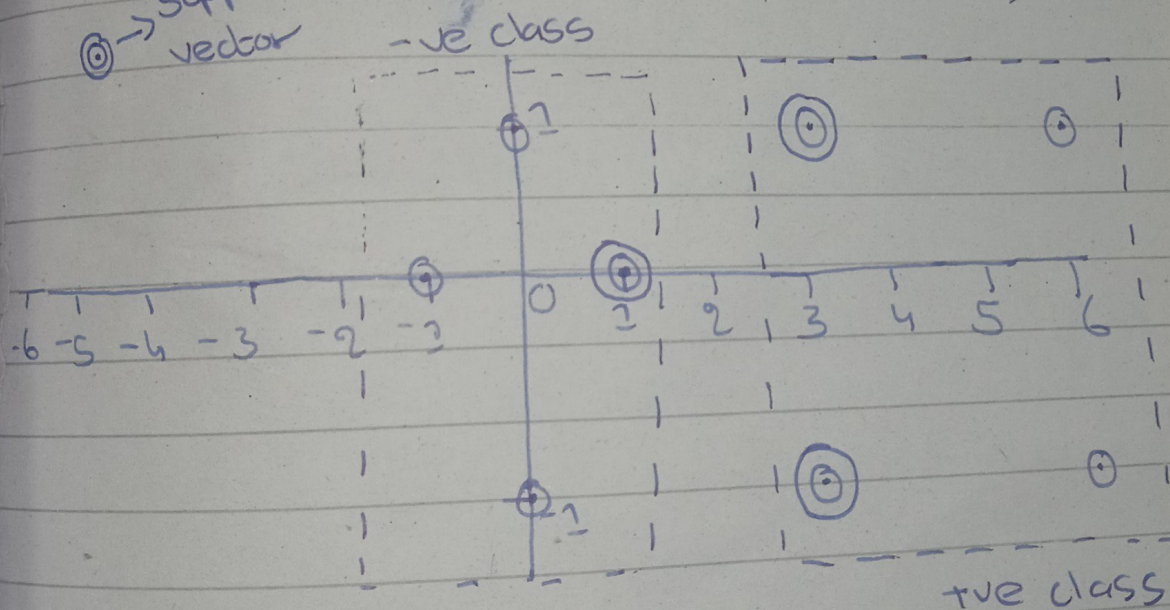
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Graph Question:-

$(3, 1)$ $(3, -1)$ $(1, 2)$ $(1, -2)$ $(2, 0)$ $(0, 2)$ $(0, -2)$ $(-2, 0)$

⊙ → support vector



$$S_1 = (1, 0), S_2 = (3, 1), S_3 = (3, -1)$$

Add bias = 1

$$S_1 = (1, 0, 1) \rightarrow \alpha_1$$

$$S_2 = (3, 1, 1) \rightarrow \alpha_2$$

$$S_3 = (3, -1, 1) \rightarrow \alpha_3$$

Now make the equations;

$$\alpha_1 S_1 S_1 + \alpha_2 S_2 S_1 + \alpha_3 S_3 S_1 = -1$$

$$\alpha_1 S_1 S_2 + \alpha_2 S_2 S_2 + \alpha_3 S_3 S_2 = 1$$

$$\alpha_1 S_1 S_3 + \alpha_2 S_2 S_3 + \alpha_3 S_3 S_3 = 1$$

$$\alpha_1 (1, 0, 1) \cdot (1, 0, 1) + \alpha_2 (3, 1, 1) (1, 0, 1) + \alpha_3 (3, 1, 1) (1, 0, 1)$$

$$\alpha_1 (1, 0, 1) \quad \alpha_2 (3, 1, 1) \quad \alpha_3 (3, 1, 1)$$

$$\alpha_1 (1+0+1) + \alpha_2 (3+0+1) + \alpha_3 (3+0+1) = -1$$

$$2\alpha_1 + 4\alpha_2 + 4\alpha_3 = -1 \quad \text{--- (i)}$$

$$\alpha_1 (1, 0, 1) \cdot (3, 1, 1) + \alpha_2 (3, 1, 1) (3, 1, 1) + \alpha_3 (3, 1, 1) (3, 1, 1) = 1$$

$$\alpha_1 (3+0+1) + \alpha_2 (9+1+1) + \alpha_3 (9+1+1) = 1$$

$$4\alpha_1 + 11\alpha_2 + 9\alpha_3 = 1 \quad \text{--- (ii)}$$

$$\alpha_1 (1, 0, 1) (3, -1, 1) + \alpha_2 (3, 1, 1) (3, -1, 1) + \alpha_3 (3, -1, 1) (3, -1, 1)$$

(3, -1, 1)

$$\alpha_1 (3+0+1) + \alpha_2 (9-1+1) + \alpha_3 (9-1+1) = 1$$

$$4\alpha_1 + 9\alpha_2 + 9\alpha_3 = 1 \quad \text{--- (iii)}$$

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Solve these equations by calculator

$$\alpha_1 = -3.5$$

$$\alpha_2 = 0.75$$

$$\alpha_3 = 0.75$$

misclassification = Soft-Margin
 Linearly Seperate = Hard-Margin

$$w = \sum_{i=1}^n \alpha_i s_i$$

$$w = \alpha_1 s_1 + \alpha_2 s_2 + \alpha_3 s_3$$

$$w = -3.5(1, 0, 1) + 0.75(3, 1, 1) + 0.75(3, -1, 2)$$

$$w = (-3.5, 0, -3.5) + (2.25, 0.75, 0.75) + (2.25, -0.75, 0.75)$$

$$w = (1, 0, -2)$$

$\underbrace{\hspace{1cm}}_{\text{orientation of hyperplane}} \quad \underbrace{\hspace{1cm}}_{\text{Position of hyperplane}}$

$(1, 0) \rightarrow$ vertical, $(0, 1) \rightarrow$ Horizontal Position of